

확률 변수 상관계수

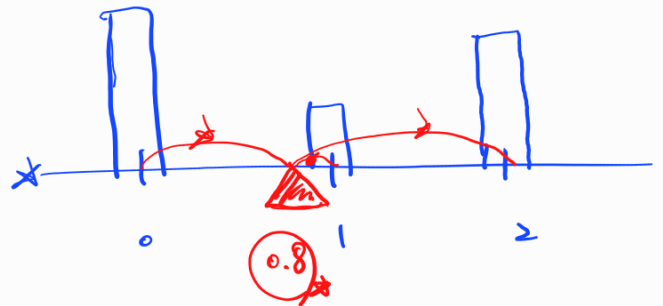
$$\text{Var}(X) = E[(X - E(X))^2]$$

$E[X]$: 확률 변수의 확률 분포의 무게 중심점.

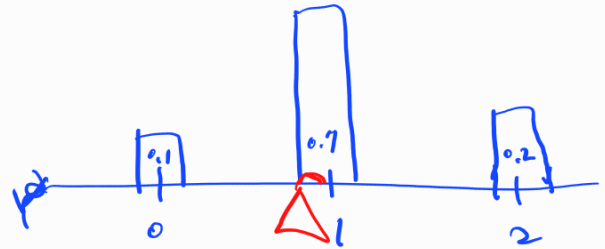
$$X = \begin{cases} 0 & 0.5 \\ 1 & 0.2 \\ 2 & 0.3 \end{cases}$$

$$0 \times 0.5 + 1 \times 0.2 + 2 \times 0.3$$

$$= 0.8$$



$$Y = \begin{cases} 0 & 0.1 \\ 1 & 0.7 \\ 2 & 0.2 \end{cases}$$



Var(Y)

0, 1, 2, 2, 1, 0, 0, 1, 1, 0, 2, 1

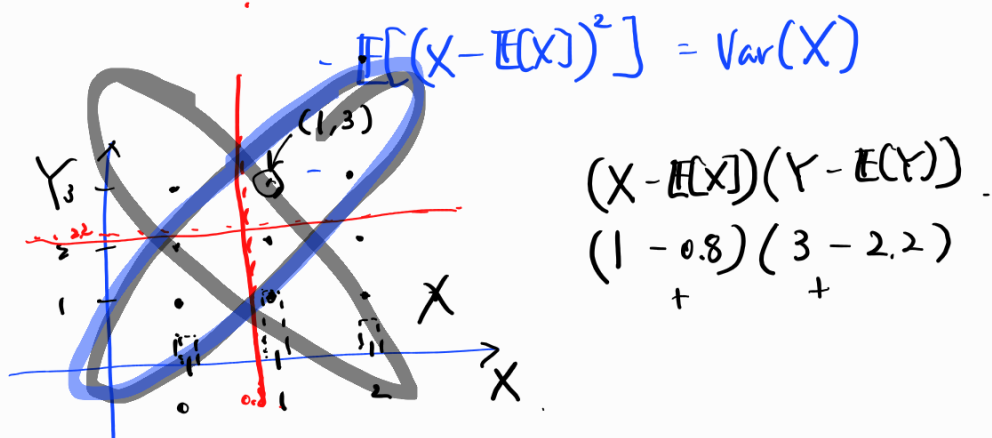
$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

* X, Y의 상관계수

$$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}$$

$$\text{Cov}(X,Y) = E[(X - E(X))(Y - E(Y))]$$

$$\text{Cov}(X,X) = E[(X - E(X))(X - E(X))]$$



$X \backslash Y$	2	4	6	$P(X=x)$
0	0.2	0	0.3	0.5
1	0.1	0.3	0.1	0.5
$P(Y=y)$	0.3	0.3	0.4	1

$\text{Cov}(X, Y)$ $E[X], E[Y], \text{Var}(X), \text{Var}(Y)$
 $= 0.5$ $= 4.2$ $= 0.25$ $= 2.76$

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}} = \frac{-0.1}{\sqrt{0.25} \sqrt{2.76}} = -0.12$$

$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$

(X, Y)	$(X - E(X))(Y - E(Y))$	P_X	$\Downarrow$
(0, 2)	$(0 - 0.5)(2 - 4.2)$	0.2	-0.1
(0, 4)	$(0 - 0.5)(4 - 4.2)$	0	
(0, 6)	$(0 - 0.5)(6 - 4.2)$	0.3	
(1, 2)		0.1	

$$\begin{pmatrix} 1,4 \\ 1,6 \end{pmatrix} \left| \begin{array}{c} \text{ } \\ \text{ } \end{array} \right. \begin{array}{c} 0,3 \\ 0,1 \end{array}$$